System of Differential Equations

$X' = AX$

$\Rightarrow D$

Put $X = PY$

$\Rightarrow Y' = DY$

$$\frac{dy}{dt} = ay$$

$$\frac{dx}{dt} = ax + by$$

$$\frac{dy}{dt} = cx + dy$$

$$X' = AX \Longrightarrow X = c_1 v_1 e^{\lambda_1 t} + c_2 v_2 e^{\lambda_2 t} + \cdots + c_n v_n e^{\lambda_n t}.$$

$$\frac{dr}{dt} = 2.5r - 1.5s \qquad r(0) = 60$$

$$\frac{ds}{dt} = -0.25r + 2.5s \qquad s(0) = 60$$

$$A = \begin{pmatrix} 2.5 & -1 \\ -0.25 & 2.5 \end{pmatrix}$$

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$\lambda = 3, 2$ $\longrightarrow \begin{pmatrix} 2 \\ 1 \end{pmatrix}$

$\downarrow$

$\begin{pmatrix} -2 \\ 1 \end{pmatrix}$

$$X = \begin{pmatrix} r(t) \\ s(t) \end{pmatrix} = c_1 \begin{pmatrix} -2 \\ 1 \end{pmatrix} e^{3t} + c_2 \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{2t}.$$

$r(0) = 60, \quad s(0) = 60$

$r(t) = -2c_1 e^{3t} + 2c_2 e^{2t}$

$s(t) = c_1 e^{3t} + c_2 e^{2t}.$

$$X = 15 e^{3t} \begin{pmatrix} -2 \\ 1 \end{pmatrix} + 45 e^{2t} \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

$$\begin{aligned} -2c_1 + 2c_2 &= 60 \\ c_1 + c_2 &= 60 \end{aligned} \qquad \begin{aligned} -c_1 + c_2 &= 30 \\ c_1 + c_2 &= 60 \end{aligned}$$

$c_1 = 15, \quad c_2 = 45$